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Graded Assignment • 20 min

Due Mar 9, 11:59 PM CET

1. Because of the large scales involved, astronomers have to use the word 'billion' (prefix: giga) a lot. So, just how big is a billion? 1 point

- ○ 10,000,000,000
- ◉ 1,000,000,000
- ○ 10,000,000
- ○ 1,000,000,000,000

2. Which branch of observational astronomy can *only* be done from space? 1 point

- ○ Optical astronomy
- ○ Infra-red astronomy
- ○ Radio astronomy
- ◉ X-ray astronomy

3. Which *three* of the following are advantages of using ground-based optical telescopes over their space-based counterparts? 1 point

- ☑ They can be repaired/upgraded more easily.
- ☑ They are comparatively cheaper.
- ☑ They can have a very large collecting area.
- ☐ They don't suffer from the effects of the Earth's atmosphere.

4. If our Sun was replaced with a similar sized star that had a hotter surface temperature, would you expect it to appear bluer or redder, and why? 1 point

- ◉ Bluer, because the hotter temperature means it will be brightest in the bluer (higher-energy) part of the spectrum.
- ○ Redder, because hot metal glows red and blue is a 'cool' colour.
- ○ Bluer, because the star will emit less red light than the Sun.
- ○ Redder, because bluer (higher-energy) light will be shifted into a part of the spectrum which we can't see.

5. If we placed the Andromeda Galaxy ten times further away from us, how much smaller and fainter would it appear? (by smaller, think angular size on the sky) 1 point

- ○ 100x smaller, 10x fainter
- ◉ 10x smaller, 100x fainter
- ○ 100x smaller, 100x fainter
- ○ 10x smaller, 1000x fainter